

Graded potentials and the action potential

These notes follow the lecture slide by slide. Each section names the slides it covers, so you can read it before the video, follow it during the video, or use it afterward to check what you remember. The three worked problems from the slides are written out in full here. Silverthorn chapter 9.

COMPETENCIES 12 TO 14 OF WEEK 4

23 SLIDES · 3 WORKED PROBLEMS · 1 MOVING MODEL

HOW TO USE THIS SHEET

Pass one, before the video: read the section, then close the page and put what you can into the box for that competency on your Competency Study Guide. Pass two, after the video: come back in a different color and add what you missed. Try each problem yourself before you read the answer.

The problem for today

SLIDES 1 TO 2 OF THE DECK

A neuron uses two kinds of electrical signal. Graded potentials are small, local and variable in size. Action potentials are large, always the same size, and able to travel the full length of an axon. In this lecture we build both from the channels that produce them. We start with graded potentials, then build the voltage gated sodium channel one gate at a time, add the voltage gated potassium channel, and use the two channels to explain every phase of the action potential. Last, we block one channel at a time and predict what happens.

Here is the problem we solve by the end. A man eats pufferfish that was not prepared correctly. Within an hour his lips and tongue are numb. Then his arms and legs grow weak, and a few hours later he cannot breathe without help. He is fully conscious the entire time. The toxin in the fish, tetrodotoxin, does one thing: it plugs voltage gated sodium channels from the outside.

Measurements from a single axon show what the toxin changes and what it leaves alone. The resting membrane potential is -70 mV before and after. Potassium inside is still 140 mM and sodium outside is still 145 mM. A weak stimulus still produces a small depolarization of about 5 mV that fades. Only one thing is different. A strong stimulus, which normally makes the membrane rise to $+30$ mV in under a millisecond, now produces a depolarization of about 20 mV that fades. Everything from the last lecture is normal, so nothing in that lecture explains this patient. Write down your best guess for why losing that one large response would stop a person from breathing.

See it move: fire an action potential yourself

SLIDE 20 OF THE DECK

This is a working model of the trigger zone in this lecture. On the left it draws the membrane potential against time. On the right is a patch of membrane with two voltage gated sodium channels and two voltage gated potassium channels, and their gates open and close in step with the tracing. Set the stimulus strength and press Stimulate, or press Step to stop at each phase and read what is happening. Keep it open and come back to it as you read the sections below. Rest in the model is the Goldman value, -71 mV, and threshold is -55 mV, so it takes a stimulus of 16 mV or more to reach threshold.

Moving model: run it on the course site, [drsrennie-stack.github.io/human-physiology-Fa26/biol005-w04-action-potential-notes.html](https://stack.github.io/human-physiology-Fa26/biol005-w04-action-potential-notes.html)

THREE THINGS TO TRY

- 1 Stimulate at 8 mV, 15 mV, 16 mV and 25 mV, and write down the highest and lowest values each time. Below threshold the response grows with the stimulus. At 16 mV and at 25 mV the highest value is $+30$ mV and the lowest is -84 mV both times.
- 2 Set the stimulus to 20 mV and use the Step button. At the peak and at the bottom of the after-hyperpolarization, read the two permeabilities and the Goldman value, and compare them with worked problem 2.
- 3 Turn on the tetrodotoxin button and stimulate at 25 mV. Then turn it off, block the voltage gated potassium channels, and stimulate again. Compare both tracings with your predictions in the section on changing the conditions.

Two kinds of electrical signal

SLIDE 3 OF THE DECK

Both signals are changes in the membrane potential away from rest, and both are produced by ion channels opening. They differ in which channels open and in what happens next. A graded potential is a small change in membrane potential whose size depends on the size of the stimulus. It happens in the input zone of the neuron, the dendrites and cell body, and it fades within a millimeter or two. An action potential is a large, brief depolarization that is the same size every time. It starts at the trigger zone, which in most neurons is the first part of the axon, and it reaches the axon terminals at full size.

The two work together. Graded potentials carry information into the trigger zone, and the trigger zone decides whether to send an action potential down the axon.

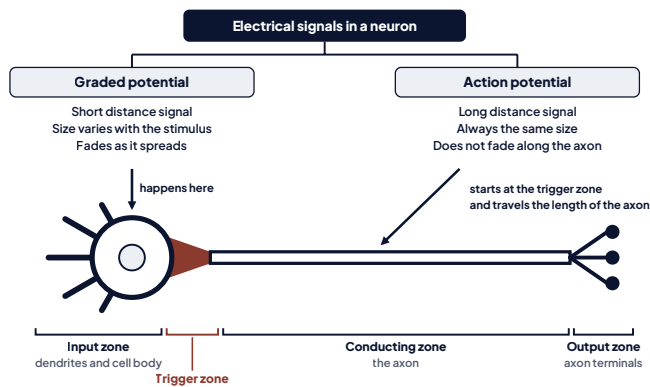


Figure 1. Graded potentials happen in the input zone. Action potentials start at the trigger zone and travel along the axon.

Graded potential

A change in membrane potential whose size varies with the strength of the stimulus and which fades with distance.

Action potential

A large, brief, all or none depolarization produced by voltage gated channels, which travels along an axon without getting smaller.

Trigger zone

The region of a neuron with a very high density of voltage gated Na^+ channels, where action potentials start. In most neurons it is the axon hillock and the first part of the axon.

Graded potentials

SLIDES 4 TO 5 OF THE DECK

A graded potential begins when a stimulus opens gated channels. At a synapse, a neurotransmitter opens ligand gated channels. In a touch receptor, stretch of the membrane opens mechanically gated channels. If the channels that open let sodium in, the membrane depolarizes. If they let potassium out or chloride in, the membrane hyperpolarizes. The rule from the last lecture still holds: the membrane potential moves toward the equilibrium potential of the ion whose permeability went up.

A stronger stimulus opens more channels, or holds them open longer, so more ions cross and the change in voltage is larger. That is what graded means. A graded potential has no threshold and no fixed size. It can be 1 mV or 20 mV, and any stimulus that opens channels produces one.

A graded potential gets smaller as it spreads. The sodium that enters is positive charge, and it spreads along the inside of the dendrite in both directions. This spread of charge is called local current flow. Two things reduce it. The first is current leak. The membrane is not a perfect insulator. It has open leak channels along its whole length, and some of the positive charge leaves the cell through them before it gets very far. The second is cytoplasmic resistance. The cytoplasm itself resists the flow of charge, in the way a thin wire resists current more than a thick one. With charge being lost through the membrane and slowed by the cytoplasm, the depolarization is smaller at every point farther from the channels that opened.

A wider dendrite has a lower cytoplasmic resistance, because there is more cross section for charge to move through, so a graded potential spreads farther in it before it fades. Widening

the tube does nothing to current leak, which depends on the leak channels in the membrane.

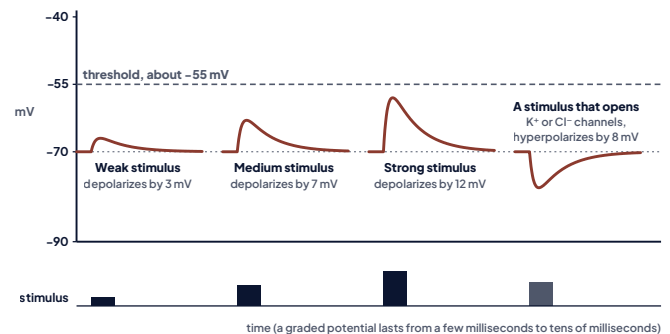


Figure 2. Three depolarizing graded potentials and one hyperpolarizing graded potential, recorded at the same spot.

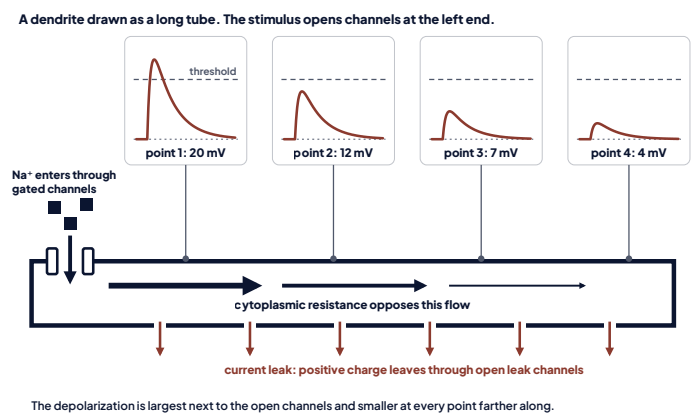


Figure 3. The same graded potential recorded at four points along a dendrite.

Local current flow

The spread of charge through the cytoplasm away from the place where ions entered.

Current leak

Loss of charge across the membrane through open leak channels as a graded potential spreads.

Cytoplasmic resistance

The resistance of the cytoplasm to the flow of charge along the inside of a dendrite or axon. It is lower in a wider process.

What reaches the trigger zone

SLIDE 6 OF THE DECK

Whether a neuron fires depends on how large the graded potential still is when it arrives at the trigger zone. A graded potential that is still large enough to reach threshold there is called suprathreshold. One that is too small is called subthreshold, and it fades without producing an action potential.

Graded potentials that overlap in time add together, which is called summation. Depolarizing graded potentials add, and hyperpolarizing graded potentials subtract. The trigger zone responds to the total.

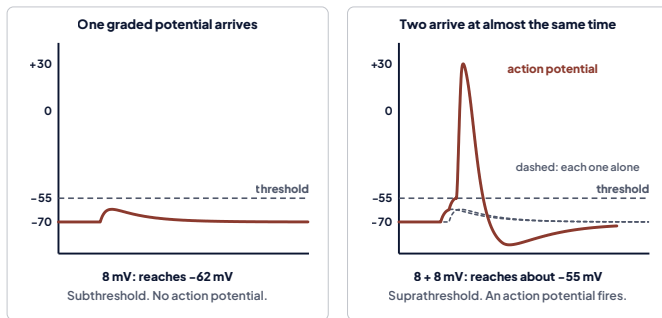


Figure 4. A graded potential that is below threshold at the trigger zone fades. Two that arrive together can add up to threshold.

WORKED PROBLEM 1

A neuron rests at -70 mV with threshold at -55 mV. One excitatory synapse produces 20 mV at the dendrite, and 8 mV of it reaches the trigger zone. Does the neuron fire with one such synapse active? With two at once? With two, plus an inhibitory synapse worth -5 mV at the trigger zone?

Work out the membrane potential at the trigger zone in each case and compare it with threshold.

THE WORK

- 1** Start from rest. The trigger zone is at -70 mV and threshold is about -55 mV, so the membrane has to depolarize by **15 mV** to reach threshold.
- 2** One synapse alone. The graded potential was 20 mV at the dendrite, but only 8 mV of it is left at the trigger zone. $-70 + 8 = -62 \text{ mV}$. That is 7 mV short of threshold, so this graded potential is **subthreshold** and no action potential fires. The membrane returns to -70 mV.
- 3** Two synapses together. Graded potentials add, so the trigger zone receives $8 + 8 = 16 \text{ mV}$. $-70 + 16 = -54 \text{ mV}$. That is past threshold, so the sum is **suprathreshold** and the neuron fires one action potential.
- 4** Now add the inhibitory synapse, which by itself hyperpolarizes the trigger zone by 5 mV. $8 + 8 - 5 = 11 \text{ mV}$. $-70 + 11 = -59 \text{ mV}$. The neuron is below threshold again and does not fire.
- 5** What the three answers say together: the neuron adds up every graded potential that reaches the trigger zone, depolarizing and hyperpolarizing, and the only question that matters is whether the total reaches threshold.
- 6** Adding graded potentials as plain numbers is a simplification. Real graded potentials that overlap add to slightly less than their arithmetic sum. The reasoning is the same.

Subthreshold

A graded potential too small to bring the trigger zone to threshold.

Suprathreshold

A graded potential large enough to bring the trigger zone to threshold.

Summation

The adding together of graded potentials that overlap in time.

The voltage gated sodium channel and its two gates

SLIDES 7 TO 10 OF THE DECK

The action potential depends on a channel that was not needed for the resting potential. A voltage gated sodium channel is a pore that only sodium fits through, and part of the protein senses the membrane potential. It has two gates in series, so sodium can pass only when both gates are open. It helps to picture a hallway with a door at each end. You can walk through only when both doors are open at the same time.

Step 1, the resting state. At the resting potential the activation gate is closed and the inactivation gate is open. One closed gate is enough to stop all sodium, so the channel is closed. This closed channel is able to open. It is waiting for a depolarization.

Step 2, the activated state. When the membrane depolarizes, the voltage sensor moves and the activation gate opens within a fraction of a millisecond. The inactivation gate was already open, so now both gates are open and sodium enters the cell. Sodium has a large driving force. At -55 mV the membrane is 116 mV away from the sodium equilibrium potential, and the concentration gradient and the voltage both point inward. The more the membrane depolarizes, the more of these channels open.

Step 3, the inactivated state. Depolarization sends the same signal to both gates. The activation gate responds quickly and opens. The inactivation gate responds slowly and closes, about half a millisecond later. So the channel is open only for the time between the fast gate opening and the slow gate closing. Once the inactivation gate has closed, the channel is inactivated. It is closed, but it is not the same as the resting state: the activation gate is open and the inactivation gate is blocking the pore. An inactivated channel cannot reopen, even if the membrane is still depolarized or is depolarized again. The channel shuts itself off, and that is what ends sodium entry.

Step 4, recovery. When the membrane repolarizes toward -70 mV, both gates return to their resting positions. The activation gate closes and the inactivation gate reopens, and the channel is closed and able to open again. So the channel has three states, closed, open and inactivated, and it moves through them in that order. It matters which gate is closed. In the resting state a depolarization will open the channel. In the inactivated state it will not, and only time spent near the resting potential resets it. A membrane whose sodium channels are inactivated cannot fire, however strongly it is stimulated. You saw this in the crush injury patient of the last lecture, and in the next lecture it explains the refractory period.

Many drugs act on this channel. Local anesthetics such as lidocaine block it, so the axons near the injection cannot carry action potentials and no pain signal reaches the brain. Some antiarrhythmic drugs and some antiseizure drugs, such as phenytoin and carbamazepine, also act on voltage gated sodium channels.

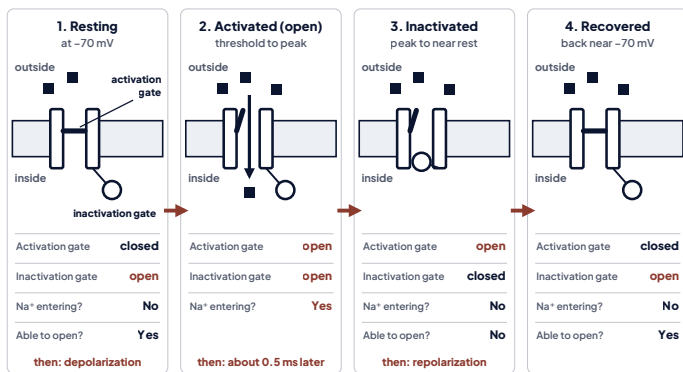


Figure 5. One channel, four drawings, three states. The bar is the activation gate and the ball on a chain is the inactivation gate. The recovered channel is back in the resting state.

Voltage gated Na⁺ channel

A sodium channel that opens when the membrane depolarizes. It has an activation gate and an inactivation gate.

Activation gate

Closed at rest. Opens within a fraction of a millisecond when the membrane depolarizes. Closes when the membrane repolarizes.

Inactivation gate

Open at rest. Closes about half a millisecond after depolarization. Reopens only after the membrane has repolarized.

Inactivated

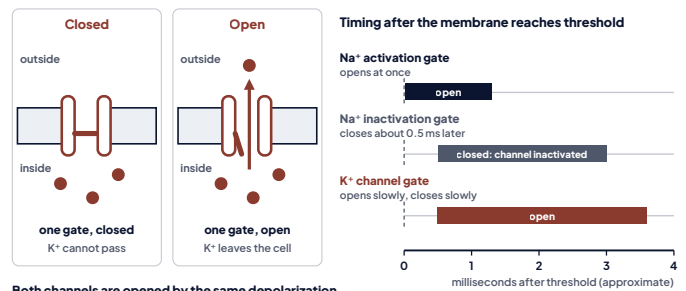
The state in which the inactivation gate blocks the pore. The channel is closed and cannot be opened by depolarization.

The voltage gated potassium channel

SLIDE 11 OF THE DECK

The second channel of the action potential is a voltage gated potassium channel. It is a different protein from the potassium leak channel that sets the resting potential. It has a single gate, which is closed at rest. The same depolarization that opens the sodium channels also opens these, but this gate is slow. It opens about when the sodium channels are inactivating. When it opens, potassium leaves the cell, because at any potential above -95 mV the net force on potassium is outward. The gate is also slow to close. It stays open until the membrane is back near rest, and for a little while after that.

The delay is essential. If the potassium channel were as fast as the sodium channel, sodium entering and potassium leaving at the same moment would cancel each other, and the membrane would hardly depolarize at all. The delay gives sodium about half a millisecond with no opposition, which is what allows the membrane to rise all the way to +30 mV. Then potassium leaves with no opposition, because the sodium channels have inactivated.



Both channels are opened by the same depolarization. The sodium channel responds in a fraction of a millisecond. The potassium channel responds about half a millisecond later, at about the time the sodium channels are inactivating.

Figure 6. The voltage gated K⁺ channel, and the timing of all three gates after the membrane reaches threshold. The times are approximate.

Threshold and the positive feedback loop

SLIDES 12 TO 13 OF THE DECK

The trigger zone has a very high density of voltage gated sodium channels. A small depolarization opens a few of them and a larger one opens more. Two currents then compete. Sodium entering depolarizes the membrane. Potassium leaving through the leak channels repolarizes it, and that outward current also grows as the membrane moves away from the potassium equilibrium potential. Below threshold the potassium current is larger, so the membrane returns to rest, and the response is only a graded potential.

Threshold is the membrane potential at which the inward sodium current just exceeds the outward potassium current. In many neurons it is near -55 mV. Books give values from about -50 to -55 mV, and it varies from cell to cell. Because threshold is a balance between two currents, it is not a fixed property of the cell. Anything that reduces the number of available sodium channels, or increases potassium permeability, moves threshold farther from rest and makes the cell harder to fire.

Past threshold there is a loop. Depolarization opens sodium channels, sodium entry depolarizes the membrane further, and that opens more sodium channels. This is positive feedback: the response increases the stimulus that caused it. It is why the rising phase is so steep, and why it cannot stop partway. Positive feedback needs something outside the loop to stop it. Here there are two things, and both were started by the same depolarization about half a millisecond earlier. Inactivation gates close, which stops sodium entry. Voltage gated potassium channels open, which lets potassium leave.

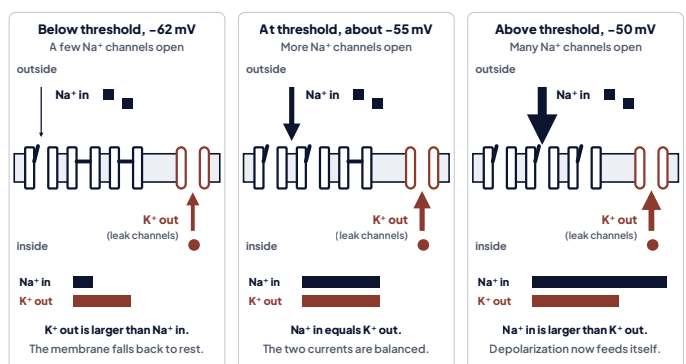


Figure 7. The trigger zone at three potentials. The bars compare the two currents.

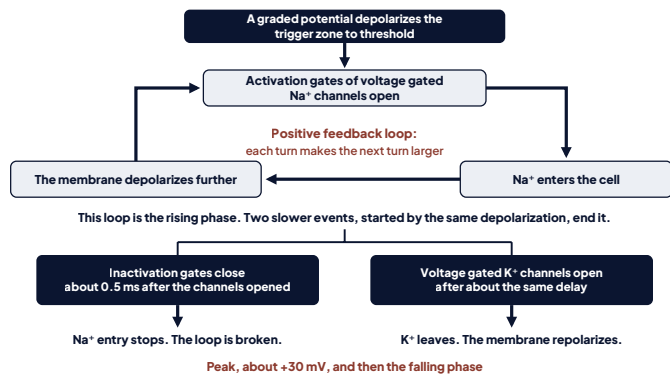


Figure 8. The positive feedback loop of the rising phase, and the two slower events that end it.

Threshold

The membrane potential at which Na^+ entry through voltage gated channels exceeds K^+ exit, so that depolarization sustains itself. About -55 mV in many neurons.

Positive feedback

A loop in which the response increases the stimulus that caused it. It continues until something outside the loop stops it.

The phases of the action potential

SLIDES 14 TO 16 OF THE DECK

1. Rest. Only leak channels are open. Potassium permeability is about 25 times sodium permeability, and the membrane sits near -70 mV . The voltage gated sodium channels are closed and able to open, and the voltage gated potassium channels are closed.
2. Depolarization to threshold. A graded potential arrives at the trigger zone and brings it to about -55 mV .
3. Rising phase. Activation gates open, sodium permeability rises several hundred times, and sodium enters. The membrane moves toward the sodium equilibrium potential, passes 0 mV and becomes positive inside. This takes well under a millisecond.
4. Peak, about $+30 \text{ mV}$. Inactivation gates are closing and potassium channels are opening, so the rise stops. The membrane never reaches $+61 \text{ mV}$. It would need the sodium channels to stay open, and they do not. Sodium entry also slows as the membrane approaches the sodium equilibrium potential, because the driving force on sodium shrinks.
5. Falling phase. Sodium channels are inactivated and voltage gated potassium channels are open. Potassium leaves, carrying positive charge out, and the membrane repolarizes.
6. After-hyperpolarization, also called the undershoot. The potassium channels are slow to close. For a few milliseconds potassium permeability is higher than at rest, so the membrane goes below -70 mV and approaches the potassium equilibrium potential. During this time the sodium channels recover: their activation gates close and their inactivation gates reopen.
7. Return to rest. The voltage gated potassium channels finish closing. Only the leak channels are open again, and they bring the membrane back to -70 mV . The sodium potassium pump does not do this. The pump is far too slow to shape a signal that lasts two milliseconds, and so few ions crossed that there is almost nothing for it to correct.

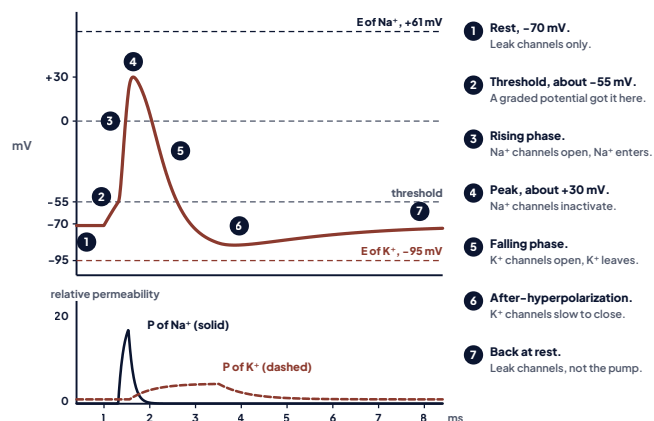


Figure 9. One action potential with the permeability to Na^+ and K^+ lined up under it.

WORKED PROBLEM 2

Use the Goldman equation, read in words, to predict where the membrane potential sits at four moments: at rest ($P_{Na} : P_K = 0.04 : 1$), with every voltage gated Na^+ channel open ($20 : 1$), at the real peak (about $4.3 : 1$), and during the after-hyperpolarization ($0.04 : 3$). For each, say whether it lies nearer E_K (about -95 mV) or E_{Na} (about $+61$ mV), and why.

$V_m = 61.5 \times \log_{10} \left(\frac{P_K [K^+]_{out} + P_{Na} [Na^+]_{out}}{P_K [K^+]_{in} + P_{Na} [Na^+]_{in}} \right)$. K^+ 4 out and 140 in. Na^+ 145 out and 15 in. The equation describes a membrane that has had time to settle. At the peak and at the bottom of the after-hyperpolarization the potential has stopped changing for an instant, so it applies exactly at those two moments.

THE WORK

- 1** At rest, $P_{Na} : P_K = 0.04 : 1$. Potassium has about 25 times the weight of sodium, so the membrane sits close to E_K , at about **-71 mV**. You met this value last lecture.
- 2** Every voltage gated Na^+ channel open, $P_{Na} : P_K = 20 : 1$. Now sodium has 20 times the weight of potassium, so the membrane swings most of the way to E_{Na} , to about **+50 mV**. This is where the membrane would settle if the sodium channels stayed open.
- 3** At the real peak, $P_{Na} : P_K$ is only about **4.3 : 1**, because many sodium channels have already inactivated and potassium channels have started to open. Sodium still has the larger weight, but by less, so the membrane sits lower, at about **+30 mV**.
- 4** After-hyperpolarization, $P_{Na} : P_K = 0.04 : 3$. Sodium permeability is back to its resting value, and potassium permeability is about three times its resting value because the voltage gated K^+ channels have not closed yet. Potassium's weight is now even larger than at rest, so the membrane sits closer to E_K than rest does, at about **-84 mV**.
- 5** Read the four answers in order: -71, toward +50, +30, -84, and then back to -71 when the potassium channels close. That is the whole shape of the action potential, and the concentrations never changed. Only the permeabilities did.
- 6** The ratios 20 to 1, 4.3 to 1 and 3 times resting are round teaching values. Measured values differ from one kind of neuron to another.

After-hyperpolarization (undershoot)

The few milliseconds after an action potential when the membrane is more negative than rest, because voltage gated K^+ channels are still open.

All or none

SLIDE 17 OF THE DECK

A stimulus that does not reach threshold gives a graded potential and no action potential. A stimulus that reaches threshold gives a full action potential. There is nothing in between. This is the all or none property. In Figure 10 the fourth stimulus is much stronger than the third, and the two action potentials have the same peak and the same shape.

The reason is that the stimulus only has to bring the trigger zone to threshold. After that the positive feedback loop runs on the energy stored in the sodium gradient, and the stimulus no longer contributes. How high the membrane goes depends on the sodium equilibrium potential, which comes from the concentrations, and on how quickly the sodium channels inactivate and the potassium channels open, which are properties of the channel proteins. None of those depends on the stimulus. Striking a match is a fair comparison. Once the match head reaches its ignition temperature it burns with the energy stored in the match, and striking it harder does not make a bigger flame.

If amplitude cannot carry information about stimulus strength, something else must. That is the first topic of the next lecture.

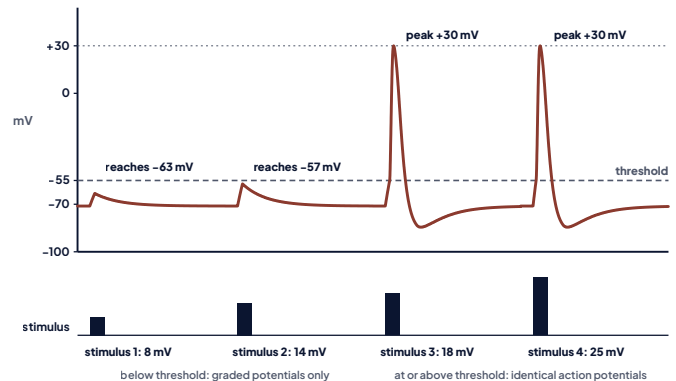


Figure 10. Four stimuli and four responses. The third and fourth responses are identical.

All or none

An action potential either happens at full size or does not happen. Its amplitude does not depend on the strength of the stimulus.

How few ions move

SLIDE 18 OF THE DECK

In the last lecture you learned that almost no ions have to move to set up the resting potential. The same is true of an action potential. The charge that has to cross the membrane to change its potential depends on the area of the membrane, and for all cell membranes it is about 6,000 single charges per square micrometer for a change of 100 mV. That figure comes from the electrical capacitance of cell membranes, about 1 microfarad per square centimeter.

Rising phase

The rapid depolarization from threshold to the peak, produced by Na^+ entering through voltage gated Na^+ channels.

Falling phase

The repolarization from the peak back toward rest, produced by Na^+ channel inactivation and K^+ leaving through voltage gated K^+ channels.

Because so few ions cross, the concentrations inside and outside the cell are the same after an action potential as before it, and the equilibrium potentials have not moved. That is why the pump is not needed to end an action potential, and why a neuron whose pumps were stopped could still fire a thousand or more action potentials before its gradients ran down.

WORKED PROBLEM 3

Take a neuron cell body as a sphere 20 μm across. How many Na⁺ ions must enter to take the membrane from -70 to +30 mV? What fraction of the sodium already inside is that?

Radius 10 μm. About 6,000 charges per μm² for 100 mV. Sodium inside is 15 mM. One mole is 6.02 × 10²³ ions.

THE WORK

- 1 Surface area of a sphere is 4πr². With r = 10 μm, the area is 4 × 3.14 × 100 = **1,257 μm²**.
- 2 Charges needed: 1,257 μm² × 6,000 per μm² = about **7.5 million Na⁺ ions** to carry the membrane from -70 to +30 mV.
- 3 Now count the sodium ions already inside. The volume of the sphere is (4 ÷ 3)πr³ = 4,189 μm³, which is 4.19 × 10⁻¹² liters. At 15 mM that is 6.3 × 10⁻¹⁴ moles, or about **38 billion Na⁺ ions**.
- 4 Compare them. 7.5 million ÷ 38 billion is about 1 in 5,000, or **0.02 percent**. Sodium inside the cell goes from 15 mM to about 15.003 mM. E_{Na} does not measurably change.
- 5 The same number of K⁺ ions leaves during the falling phase. The cell holds about 350 billion K⁺ ions, so it loses roughly 1 in every 47,000.
- 6 A real neuron moves somewhat more than this minimum, because for part of each action potential sodium is entering while potassium is leaving. The conclusion is the same: one action potential does not change the concentrations, so the pump is not needed to end it. The pump matters over thousands of action potentials, when it returns the small amounts that crossed.

The two signals side by side

SLIDE 19 OF THE DECK

The graded potential is how a neuron measures its input. Its size reports how strong the stimulus was, and summation lets the neuron combine many inputs. But it fades, so it cannot carry a message farther than a millimeter or two. The action potential is how the neuron sends its decision over a distance. It does not fade, because it is regenerated by voltage gated channels all along the axon, and the cost of that is a fixed size.

GRADED POTENTIAL AND ACTION POTENTIAL COMPARED

	GRADED POTENTIAL	ACTION POTENTIAL
Where it happens	Input zone: dendrites and cell body. Also sensory receptor endings.	Starts at the trigger zone and travels along the axon
Channels that produce it	Ligand gated or mechanically gated channels	Voltage gated Na ⁺ and K ⁺ channels
Direction of the change	Depolarizing or hyperpolarizing	Always a depolarization, followed by repolarization and a brief after-hyperpolarization
Size	Varies with the strength of the stimulus	All or none. Always about 100 mV.
Over distance	Gets smaller, because of current leak and cytoplasmic resistance	Arrives at the axon terminal at full size
Threshold	None	Requires a suprathreshold graded potential at the trigger zone
Can two add together?	Yes	No
Refractory period	None	Yes. Covered in the next lecture.
Its job	To carry the strength of the input to the trigger zone	To carry a signal over a long distance without loss

Changing the conditions

SLIDE 21 OF THE DECK

Block the voltage gated potassium channels and leave everything else normal. The rising phase and the peak are normal, because they depend on sodium channels. The falling phase is much slower. Sodium channels still inactivate, so sodium entry stops, but the only way for potassium to leave is the leak channels. The action potential lasts several times longer and there is no after-hyperpolarization. Tetraethylammonium, a laboratory potassium channel blocker, does this.

Stop the inactivation gate from closing. The membrane rises normally and then stays depolarized, because sodium keeps entering while potassium leaves. Repolarization is delayed and the action potential is greatly prolonged. Some scorpion and sea anemone toxins work this way. The experiment shows what inactivation is for: it is the step that ends sodium entry.

Block the voltage gated sodium channels with tetrodotoxin. The stimulus still depolarizes the membrane, because the channels that produce graded potentials are not voltage gated sodium channels. The depolarization passes -55 mV and nothing happens, because the channels that would start the positive feedback loop cannot conduct. There is no rising phase, so there is no action potential at any stimulus strength.

The patient, explained

SLIDE 22 OF THE DECK

Tetrodotoxin plugs the outer mouth of the voltage gated sodium channel. It does not touch leak channels, the pump, or ligand gated channels, so the diner's resting potentials are normal and his graded potentials are normal. What he has lost is the rising phase. With the sodium channels blocked, reaching threshold

opens nothing. There is no positive feedback loop, so no action potential forms in the axon.

Sensation fails first. Numb lips and tongue come first, because sensory axons can no longer carry action potentials. Then motor neurons can no longer carry commands to skeletal muscle, and skeletal muscle fibers, which use the same kind of sodium channel, cannot fire either. When the diaphragm stops receiving action potentials from the phrenic nerve, breathing stops. His heart keeps beating because heart muscle uses a different voltage gated sodium channel that is far less sensitive to tetrodotoxin. The toxin also crosses poorly from the blood into the brain, so the neurons of his brain keep working, and he is conscious while unable to move.

Look at it as an experiment. The problem is that no action potentials reach the diaphragm or the other breathing muscles, so air does not move. The target variable is ventilation of the lungs, and through it blood oxygen and carbon dioxide. The intervention is a ventilator, which moves the air that the muscles cannot. It does nothing to the toxin or to the channels. Recovery comes from the fact that tetrodotoxin binding is reversible. As the body clears the toxin over hours to days, it leaves the channels, and they work normally again, because they were never damaged. With ventilation maintained, strength and sensation return as the toxin is cleared.

Local anesthetics block the same channel from the inside of the membrane. The numb lip after dental work and the numb lips of this patient are the same physiology at very different scales.

What this lecture assumes you already have

If any line here is shaky, fix it before you work the lecture.

CHEMISTRY

What an ion is, and the charges on Na^+ and K^+
Opposite charges attract and like charges repel
Diffusion: particles move from higher concentration toward lower

Chemistry review page

FROM EARLIER LECTURES

The resting membrane potential: E_K is -95 mV , E_{Na} is $+61 \text{ mV}$, and rest is -70 mV
The Goldman equation, and that raising the permeability to an ion moves the membrane potential toward that ion's equilibrium potential
Leak, voltage gated, ligand gated and mechanically gated channels
Depolarization, repolarization, hyperpolarization and overshoot
The four zones of a neuron: input zone, trigger zone, conducting zone and output zone

No new anatomy is needed for this lecture, so there is no anatomy list this time.

MATH

Scientific notation, such as 6.02×10^{23}
Area and volume of a sphere (the formulas are given in the problem)

Close the page and produce it

These are the Week 4 competencies this lecture covers, straight off the study guide, with the two prompts that go with each one. Pick one prompt per competency. Blank paper, nothing open.

Read the full instructions first, including how the first pass before the lecture differs from the second pass after it.

12

Graded potentials

Describe how a graded potential is produced and explain why it varies with stimulus strength and decays with distance, using the terms current leak and cytoplasmic resistance, and relate current, voltage, and resistance with Ohm's law, including how opening channels raises conductance.

Do one of the two. Head your paper with the number.

12-1

Draw a dendrite as a long tube with a stimulus at one end. At four points along it, draw the membrane potential as a small trace, showing the signal shrinking as you move away. Draw current leaking out through channels along the way and label the cytoplasmic resistance inside the tube. One line: name the two reasons the signal shrank, and say which one you could reduce by making the tube wider.

12-2

Draw one set of axes, membrane potential against time, and plot three graded potentials from the same spot produced by a weak, a medium, and a strong stimulus. Then plot two of them arriving close together and summing. Mark threshold as a dashed line. One line: a graded potential changes size with stimulus strength and an action potential does not, so say what the graded potential's actual job is.

13

Action potential phases

Diagram an action potential and state the channel state and ion movement responsible for depolarization to peak and for repolarization and afterhyperpolarization.

Do one of the two. Head your paper with the number.

13-1

Draw an action potential tracing, mV on the y and ms on the x, from minus 70 up to about plus 30 and back down through an afterhyperpolarization near minus 80. Label threshold, depolarization, peak, repolarization, and afterhyperpolarization on the curve. Underneath, draw two permeability traces lined up in time with it, one for sodium and one for potassium. One line: say what closes at the peak and what is still open during the afterhyperpolarization.

13-2

Draw a voltage gated sodium channel four times across the top of your box, in its resting, activated, inactivated, and recovered states, with both of its gates drawn separately every time. Under each, write the membrane voltage at which that state occurs. Beneath that row, draw a voltage gated potassium channel in its two states with its timing. One line: both open on depolarization, so say what makes potassium arrive late and why the cell needs that delay.

14

Threshold and all or none

Explain what threshold represents at the trigger zone and why action potential amplitude does not change with a stronger stimulus.

Do one of the two. Head your paper with the number.

14-1

Draw the trigger zone as a close up patch of membrane with voltage gated sodium channels in it, three times over: below threshold with a few channels open and the leak winning, exactly at threshold with the two currents balanced, and above threshold with the positive feedback running away. Draw the current arrows in and out in every panel. One line: define threshold from your three drawings in terms of two currents, not a number.

14-2

Draw one set of axes with four stimuli of increasing strength as bars on a bottom trace and the resulting membrane potential plotted above them. Show the first two failing to reach threshold and the last two producing identical full sized action potentials. Mark threshold. One line: the fourth stimulus was far stronger than the third and the spike came out the same size, so say what actually fixes the amplitude.

Dr. Sharilyn Rennie · BIO 005 Human Physiology · Graded potentials and the action potential. Concentrations and permeability ratios are typical teaching values. All figures and the moving model are original drawings made for this course.